In-Context Gibbs: Learning from Weak Labels with Frozen Tabular Foundation Models

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Abstract

Tabular foundation models (TFMs) predict labels in context without training, but they need labelled examples. We show that a frozen TFM can learn from weak labels, such as candidate sets or label proportions within bags, without any retraining. Under exchangeability, the distribution of one unknown label given all the others is a posterior predictive with the other points as context, which is what a TFM computes. In-Context Gibbs therefore samples the latent labels with forward passes of the TFM, brings in the weak labels through their likelihood, and predicts by averaging over the samples. We prove that, with an exact TFM, its exact variants target the Bayes posterior, and that the sampler remains well defined and stable when the TFM's conditionals are approximate or even mutually incompatible. In a pre-registered study on 28 new datasets, In-Context Gibbs with TabICL v2 outperformed the strongest specialised baselines both for learning from label proportions (+0.052 AUC over DLLP, better on 16 of 18 datasets) and from partial labels (better than tuned CC at both ambiguity levels), and it had the highest accuracy on three real partial-label datasets. An earlier pre-registered study with TabPFN v2, in which the same sampler stalled under heavy ambiguity, identifies the key requirement: the TFM must be robust to noisy context labels.

1 Introduction

Labels in tabular data often arrive weak: an annotator who cannot decide records a *candidate set*, and data released under privacy constraints give only the *proportion* of each label within groups. Each such form of weak supervision has its own family of learners, trained per task with a loss designed for that observation and tuned on validation splits that, too, carry only weak labels [2, 6, 10, 20, 29].

Tabular foundation models (TFMs) have changed what is possible on small tables with clean labels. TabPFN [13, 14] and TabICL [23, 24] predict the label of a query point in a single forward pass, conditioning on a labelled context, without any training or tuning. Weak labels do not fit this interface: the context must be labelled. The obvious workaround, filling the context with one plausible label per point, throws away much of what the weak labels say: for candidate sets and label proportions, the guessed labels carry only a fraction of the information in the observation (Proposition 2).

A TFM is already a Gibbs sampler for unknown labels. A TFM approximates the posterior predictive distribution (PPD) of its prior, $p(y_* \mid x_*, D)$ [22]. Under the exchangeability built into that prior, the conditional distribution of one unknown label given all the others is precisely such a PPD, with the other points and their current labels as the context (Lemma 1). A forward pass of a frozen TFM is therefore the full conditional of a Gibbs sampler over the latent labels, and weak labels enter only through their likelihood. In-Context Gibbs exploits this: it samples the unknown labels block by block with forward passes, respecting candidate sets or bag counts, and predicts by averaging the TFM's predictions over the samples (Section 4). No weight is updated, and the same procedure handles instance-level and group-level weak labels.

Evidence from two pre-registered studies. We evaluated the method in two studies whose hypotheses, comparators and tests were fixed before the evaluation data were touched. Study 1, with TabPFN v2, showed large gains over the naive use of the TFM and located what limits them: on partial labels with high ambiguity and many classes, TabPFN v2 reinforces an uninformative labelling, and the chains stall. In the registered TFM ablation with TabICL v2, which pulls such labellings back towards the truth, the same sampler passed every test. Study 2 put this to a prospective test on 28 new datasets, changing only the TFM (Table 2). In-Context Gibbs outperformed the strongest specialised baselines for learning from label proportions, by +0.052 AUC over DLLP on 16 of 18 datasets, and from partial labels, by about one accuracy point over tuned CC at both ambiguity levels, with no collapse on datasets with many classes. On three real partial-label datasets it was the most accurate method (Section 6.5).

Contributions.

- **Method** (Section 4): In-Context Gibbs, which turns a frozen TFM into a sampler over latent labels for instance-level and group-level weak labels, at the cost of a few hundred forward passes [C2].
- **Theory** (Section 5): with an exact TFM the exact variants of the sampler target the Bayes posterior and their predictor is Bayes-optimal under the TFM's prior [C3]; with an approximate TFM, even one whose conditionals are incompatible, it has a unique stationary distribution whose error is bounded by the TFM's error times the mixing time of the exact chain [C4]; and the naive use of a TFM loses information exactly for candidate sets and bag counts [C12].
- **Evidence** (Section 6): a pre-registered confirmatory study in which In-Context Gibbs beats the strongest specialised baselines on both settings [C6][C8], supported by real partial labels [C9].
- **Mechanism** (Sections 6.3 and 6.4): the gain comes from modelling the dependence among latent labels [C7], and it is realised when the TFM is robust to noisy context labels, which a one-sweep response map makes visible [C13].

2 Related work

Learning from weak labels. Each type of weak supervision has its own family of learners, trained by gradient descent with a loss tailored to the observation: for partial labels, progressive identification of the true label [20], consistent risk estimators [10] and contrastive disambiguation [31]; for label proportions, bag-level losses [2, 4], consistency regularisation [29] and belief propagation over bags [12]. Unified frameworks treat many types with one expectation–maximisation scheme [5, 6, 28]; GLWS computes the E-step exactly with an automaton over label sequences, under the assumption that predictions are conditionally independent given the inputs. We share the goal of one procedure for many weak-label types, but replace training by inference: the latent labels are sampled jointly, and their dependence is modelled through the TFM's context rather than assumed away.

Table 1: Positioning against the closest works. "Partial": exact posterior over labels inside an EM loop, with conditionally independent predictions.

	no training (frozen model)	tabular data	several weak-label types	inference over latent labels with guarantees	dependence among latent labels
In-Context PU [19]	– (pretrains a PFN)	✓	– (PU only)	–	–
GLWS [6], ILL [5]	–	–	✓	partial	–
TabPFN imputation [27]	✓	✓	– (missing outcome)	–	–
Noise immunity [15]	✓	✓	– (label noise)	– (measurement)	–
In-Context Gibbs (ours)	✓	✓	✓	✓	✓

Tabular foundation models and in-context learning. Prior-fitted networks are trained to approximate the posterior predictive distribution of a prior over datasets [22]; TabPFN [13, 14] and TabICL [23, 24] turned this into strong off-the-shelf predictors for small tables, and recent theory studies when transformers learn posterior predictives in context [16]. Their behaviour with imperfect context labels has been measured [15] and, for unlabelled context, analysed theoretically [17]; other work pretrains new in-context models for a single weak-label type, such as positive–unlabelled data [19] or semi-supervised learning [8]. We instead keep a general TFM frozen and use it as the conditional of Bayesian inference over latent labels, for any weak label with a known observation model.

Gibbs sampling with learned conditionals. Drawing labels from TabPFN's predictive has been used for single-pass multiple imputation of a missing outcome [27], and transformers have been pretrained to impute missing features [9]. Iterative imputation with chained conditional models [18] and amortised Gibbs samplers [32] are closest in spirit; our sampler conditions every label on all others, uses the weak-label likelihood, and has guarantees even for incompatible conditionals.

3 Problem setup

We observe features $X = (x_1, \dots, x_n)$ whose labels $y \in [C]^n$ are latent. As in the prior-fitted networks behind TFMs [22], we assume an exchangeable prior: $\phi \sim p(\phi)$ and $(x_i, y_i) \overset{\text{iid}}{\sim} p(x, y \mid \phi)$, with a test pair (x_*, y_*) from the same mechanism. For a labelled context D, the *posterior predictive distribution* (PPD) of the prior is $p(y_* \mid x_*, D)$; a TFM returns an approximation $q(y_* \mid x_*, D)$ in one forward pass. Instead of y we observe weak labels W with a known likelihood $\ell(y) = p(W \mid X, y)$, independent of (x_*, y_*) given (X, y) (A0). For *partial labels* (PLL) each point has a candidate set $S_i \ni y_i$ and $\ell(y) = \prod_i \mathbb{1}[y_i \in S_i]$, under the standard assumption that the probability of a candidate set does not depend on which member is true. For *label proportions* (LLP) the points form disjoint bags B_g with observed label counts w_g, and $\ell(y) = \prod_g \mathbb{1}[\text{counts}(y_{B_g}) = w_g]$. PU data, class-dependent noise and semi-supervised learning are further instance-level cases with $\ell(y) = \prod_i \ell_i(y_i)$ (Appendix I). The targets are the posterior over latent labels and the predictive,

$$\pi(y) = p(y \mid X, W) \propto p(y \mid X)\, \ell(y), \qquad p(y_* \mid x_*, X, W) = \textstyle\sum_y \pi'(y)\, p(y_* \mid x_*, (X, y)), \quad (1)$$

with $\pi'(y) = p(y \mid x_*, X, W)$, computed without updating any weight of the TFM. The analysis assumes (A1) an exact TFM, $q = p$, on every queried context; (A2) a positive PPD; and (A3) $\sum_y p(y \mid X)\ell(y) > 0$. Theorem 2 removes (A1).

4 In-Context Gibbs

The full conditional is one forward pass. Exchangeability turns a TFM into a Gibbs kernel. For every i and every assignment y with $p(y_{-i} \mid X) > 0$,

$$p(y_i \mid y_{-i}, X) = p(y_i \mid x_i, D_{-i}), \qquad D_{-i} = \{(x_j, y_j)\}_{j \neq i} \quad (2)$$

Algorithm 1 In-Context Gibbs (block version; LLP uses bag-blocks). No TFM weight is updated.

Require: features X, weak labels W with model ℓ, frozen TFM q, test points X_*; K blocks, T sweeps, burn-in T_0, S samples

$P_0 \leftarrow$ warm start; draw $y \sim P_0 \cdot \ell$ ▷ LLP: conditioned on bag counts

for $t = 1, \dots, T$ **do**

  partition all points (LLP: all bags) at random into blocks $B_1, \dots, B_K$

  for $k = 1, \dots, K$ **do**

    $Q \leftarrow q(\cdot \mid x_i, D_{-B_k})$ for all $i \in B_k$ ▷ one forward pass

    PLL: draw $y_i \propto Q_i \, \ell_i$ for free $i \in B_k$; LLP: draw $y_{B_g} \sim \prod_{i \in B_g} Q_i$ given counts w_g, for $B_g \subseteq B_k$

    (optional) accept the new y_{B_k} with the MH probability of Proposition 3(a)

  if $t > T_0$ **then** store $y^{(t)}$

return $\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_s q(y_* \mid x_*, (X, y^{(s)}))$ over S evenly spaced stored samples

(Lemma 1): the conditional of one latent label given all the others is the PPD of a query point x_i with the rest of the current assignment as its context. Multiplying by the weak-label factor gives the full conditional of π in (1), $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i}) \, \ell_i(k)$. A Gibbs sweep over latent labels therefore needs only forward passes of a frozen TFM, and the weak labels enter only through ℓ.

Blocks. One forward pass per label would cost n passes per sweep. We instead partition all n points at random into K blocks in every sweep and, for each block B, query the TFM once with the context D_{-B} (the other blocks, with their current labels). For PLL, each free y_i, $i \in B$, is drawn from $\propto q(\cdot \mid x_i, D_{-B}) \, \ell_i(\cdot)$; blocks include the points whose label is fixed, since blocking only free points would bias the composition of the context. For LLP, blocks are groups of whole bags (*bag-blocks*), and each bag's labels are drawn from $\prod_{i \in B_g} q(y_i \mid x_i, D_{-B})$ conditioned on its counts, exactly, by dynamic programming. An optional Metropolis–Hastings (MH) correction accepts or rejects the whole block, which makes the block update exact (Proposition 3); it costs one forward pass per changed label. Contexts larger than 2,000 points are subsampled uniformly at random.

Warm start and prediction. The chain starts from a cheap estimate P_0 of the labels: for PLL, the TFM fitted on one candidate per point chosen uniformly at random, cross-fitted over 5 folds; for LLP, linear DLLP [2]. Initial labels are drawn from P_0 restricted by ℓ (for LLP, conditioned on the bag counts). After T sweeps, of which the first T_0 are discarded, the prediction for a test point is the Rao–Blackwellised average over S evenly spaced samples,

$$\hat{p}(y_* \mid x_*) = \frac{1}{S} \sum_{s=1}^{S} q(y_* \mid x_*, (X, y^{(s)})), \tag{3}$$

computed with the TFM's prediction ensemble. This *inductive* predictor is our default; a *transductive* variant, in which test points join the chain, is evaluated as an ablation.

Cost and settings. A run costs $TK + S$ forward passes and no gradient step. All settings were fixed on development datasets before any evaluation and are shared by every dataset and both studies: $K = 10$, $T = 30$, $T_0 = 10$, $S = 10$, i.e. 310 forward passes.

Comparators on the same TFM. To isolate what the sampler adds, we also run with the same TFM: the *naive TFM*, conditioned on one label per point consistent with the weak labels (a uniformly chosen candidate; a random count-consistent assignment per bag); *hard-EM*, with the same warm start and blocks but the constrained arg max instead of sampling (10 iterations); *stochastic EM*, which predicts from the last sample of the chain; and an *oracle* given the true labels.

5 Theory

We state what holds when the TFM is exact, what survives when it is not, and what the default sampler of Algorithm 1 is *not* covered by. Proofs are in Appendix A.

Lemma 1 (Full conditionals)**.** *Under the exchangeable prior,* (2) *holds for every i and every y with $p(y_{-i} \mid X) > 0$.*

Theorem 1 (Exactness)**.** *Assume (A0)–(A3) and $q = p$. (a) For instance-level weak labels, single-site Gibbs updates $\pi(y_i = k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$ satisfy detailed balance with respect to π, and the chain is irreducible and aperiodic on $\mathrm{supp}(\pi)$, hence converges geometrically. (b) For LLP, the MH chain that swaps two differently labelled points of one bag, with acceptance ratio computed from two forward passes, is reversible and irreducible on $\mathrm{supp}(\pi)$. (c) The inductive estimator* (3) *converges almost surely to $\bar{p}(y_* \mid x_*) = \sum_y \pi(y)\, p(y_* \mid x_*, (X, y))$, and $|\bar{p} - p(y_* \mid x_*, X, W)| \le \mathrm{TV}\big(p(y \mid X, W),\, p(y \mid x_*, X, W)\big)$, which is zero when the prior is* separable *(the feature distribution carries no information about $p(y \mid x)$); the transductive chain converges to $p(y_* \mid x_*, X, W)$ exactly.*

Proposition 1 (Bayes optimality under the prior)**.** *For any strictly proper scoring rule, $p(y_* \mid x_*, X, W)$ uniquely minimises the Bayes risk over all measurable predictors of (X, W, x_*), the expectation being taken under the TFM's prior.*

With an exact TFM, In-Context Gibbs thus computes the Bayes answer under the TFM's prior [C3][C10]; on real data, which need not follow that prior, the evidence is empirical. A real TFM is only approximately a PPD, and its conditionals need not be compatible with any joint distribution. The next result covers the sampler as run.

Theorem 2 (Stability under an approximate TFM)**.** *Let P be the kernel of the algorithm with the exact PPD (stationary distribution π) and $\tilde{P}$ the same algorithm with a TFM $q > 0$. Then $\tilde{P}$ has a unique stationary distribution $\tilde{\pi}$, even when the conditionals of q are incompatible, and*

$$\mathrm{TV}(\tilde{\pi}, \pi) \le \tau(P)\, \bar{\varepsilon}, \qquad \tau(P) = \sum_{t \ge 0} \delta(P^t), \quad \bar{\varepsilon} = \sum_y \tilde{\pi}(y)\, \mathrm{TV}\big(\tilde{P}(y, \cdot), P(y, \cdot)\big), \tag{4}$$

where δ is Dobrushin's ergodicity coefficient. For the random-scan single-site kernel, with $\varepsilon_{\mathrm{TV}}$ the TV error of the TFM's conditionals and κ a constant that depends only on ℓ, $\bar{\varepsilon} \le \min\{1,\, 2\kappa\, \varepsilon_{\mathrm{TV}}\}$; predictions inherit the error additively (Appendix A).

The bound reads as (integrated mixing time of the exact chain) $\times$ (error per step) [C4]. When the conditionals are incompatible, $\tilde{\pi}$ is in general not the posterior π; the theorem bounds how far it is. Its form is classical [7, 21, 25, 26] and tight in general; what is new is that a chain built from an incompatible TFM still has a unique stationary distribution, with error terms a TFM actually has, non-asymptotically and including MH steps, unlike the iterative-imputation analysis of Liu et al. [18].

Blocks and scope. A block update becomes exact when its proposal, the product of block-wise TFM predictives (for LLP, conditioned on bag counts), is accepted with the MH ratio computed by Brook's lemma along the changed labels (Proposition 3(a), Appendix A); without that step, Theorem 2 still applies, but its error term is controlled only under an extra stability assumption and grows with n for fixed K (Proposition 3(b)). The default configuration of our experiments, uncorrected blocks with $K = 10$ and bag-blocks for LLP, is therefore not covered by a non-vacuous guarantee. Section 6.6 shows that the exact sampler does not predict better, at many times the cost.

When can the sampler help? The naive use of a TFM conditions it on one label per point drawn from the weak labels; let $\tilde{\eta}_k(x) = P(\tilde{w} = k \mid x)$ be the function that a consistent learner fitted on these labels approaches, and $\eta_k(x) = p(y = k \mid x)$ the true posterior.

Proposition 2 (When the naive TFM suffices)**.** *(a) If the naive label depends on x only through y, with equal off-diagonal entries in each column of the channel, then $\tilde{\eta}_k = b_k + \lambda_k \eta_k$ with $\lambda_k > 0$, so the naive target ranks points exactly as η_k; this covers PU-SCAR, class-dependent noise, SSL with labels missing completely at random, PLL with independent distractors and LLP with random bags. (b) If the naive labels are drawn from the observation by a kernel that does not depend on the parameter, their Fisher information is at most that of the observation, with equality iff the observation's score is a function of the naive labels: equality holds for PU-SCAR and label noise, whereas for PLL and LLP the naive labels coarsen the observation (for LLP with bags of 8, about 3% of the information in the bag counts in a logistic model).*

Only for candidate sets and bag counts, then, does the naive TFM leave information that the sampler can recover through ℓ; this predicted, before evaluation, where In-Context Gibbs would beat the naive TFM [C12] (Appendix I).

6 Experiments

6.1 Design: two pre-registered studies

Settings and baselines. The main claims concern LLP with random bags of 8 (ROC AUC) and PLL in which each wrong class enters the candidate set independently with probability $q \in \{0.3, 0.6\}$ (accuracy). Weak labels are generated on the training part of stratified 70/30 splits (seeds 0–2) shared by all methods; datasets have 500 to 10,000 rows. We run the peer-reviewed baselines of each setting, each in its published default and tuned by 20 random configurations scored on a validation split *with weak labels only*: for LLP, DLLP [2] with an MLP or a linear model, LLP-VAT [29], EasyLLP [4] and GLWS [6]; for PLL, PRODEN [20], GLWS/ILL [5, 6], RC and CC [10]. In-Context Gibbs is never tuned per dataset. A rule applied to development datasets before any evaluation fixed the *primary comparators*: linear DLLP and tuned CC.

Hypotheses. H2 (LLP) and H4 (PLL) require In-Context Gibbs to beat both the primary comparator and the naive TFM (two-sided Wilcoxon signed-rank over datasets, each averaged over 3 seeds, Holm-adjusted) *and* to beat the baseline with the best mean score on $\geq$70% of datasets (LLP), or to match or beat it on $\geq$60% of datasets at each level (PLL). H3 compares In-Context Gibbs with hard-EM on log loss (Bayesian signed-rank test, ROPE ± 0.005 [3]). Every test is repeated without datasets flagged in advance for group structure or drift.

Two studies. *Study 1* froze its protocol before any evaluation data were used and evaluated 24 OpenML-CC18 datasets (10 binary, 14 multiclass) with TabPFN v2 as the registered TFM, and TabICL v2 as a registered TFM ablation. *Study 2* replicated it on 28 new datasets (18 binary, 10 multiclass) with TabICL v2, changing nothing else. Its hypotheses (H2′–H4′), comparators, tests and dataset-selection rules were fixed in a timestamped commit before any of its evaluation tables was downloaded (Appendix B); its datasets come from five curated suites [e.g. 11], excluding OpenML-CC18, Study 1 and synthetic data. Every run used one T4 GPU. Numerical checks of the theory on problems with an exact posterior are in Appendix E.

6.2 Main result: In-Context Gibbs outperforms specialised learners

In the confirmatory study all pre-registered tests passed (Table 2, right; Figure 1). On LLP with bags of 8, In-Context Gibbs improved AUC over linear DLLP by +0.052 [+0.024, +0.086] (16/0/2, $p_{\mathrm{Holm}} < 0.001$) and beat the strongest LLP baseline, tuned DLLP (MLP), on 16/18 datasets [C6]. On PLL it improved accuracy over tuned CC by +0.012 at $q = 0.3$ (8/2/0) and +0.012 at $q = 0.6$ (8/1/1;

Table 2: Pre-registered confirmatory tests. Δ: mean over datasets of (In-Context Gibbs − comparator), each dataset averaged over 3 seeds; W/T/L: datasets won/tied/lost; p_{Holm}: two-sided Wilcoxon signed-rank, Holm-adjusted within each hypothesis. Bold: significant in favour of In-Context Gibbs. "Naive TFM" uses the study's TFM. H2/H4 additionally required winning against the strongest baseline on $\geq$70%/$\geq$60% of datasets.

			Study 1: TabPFN v2			Study 2: TabICL v2, new datasets		
Setting	Metric	Gibbs vs.	Δ	W/T/L	p_{Holm}	Δ [95% CI]	W/T/L	p_{Holm}
LLP, bag size 8	AUC	DLLP (linear)	+0.024	8/0/2	0.232	+0.052 [+0.024, +0.086]	16/0/2	$<$**0.001**
		Naive TFM	+0.148	10/0/0	**0.004**	+0.116 [+0.085, +0.150]	18/0/0	$<$**0.001**
PLL, $q = 0.3$	Acc.	CC (tuned)	+0.022	10/0/4	0.084	+0.012 [+0.006, +0.019]	8/2/0	**0.016**
		Naive TFM	+0.099	13/0/1	$<$**0.001**	+0.022 [+0.012, +0.033]	10/0/0	**0.008**
PLL, $q = 0.6$	Acc.	CC (tuned)	−0.132	7/0/7	0.715	+0.012 [+0.006, +0.019]	8/1/1	**0.016**
		Naive TFM	+0.180	12/0/2	**0.002**	+0.205 [+0.094, +0.337]	10/0/0	**0.008**
Pre-registered verdict			H2: not met; H4: not met			H2′: met; H4′: met		

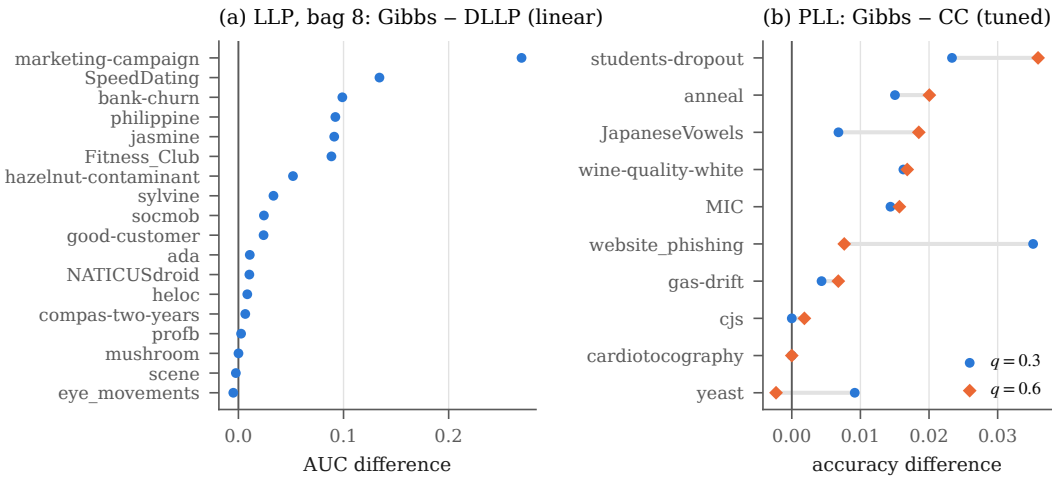


Figure 1: Confirmatory study, per dataset: In-Context Gibbs minus the primary comparator (positive favours In-Context Gibbs), mean over 3 seeds. (a) LLP with bags of 8, AUC against linear DLLP. (b) PLL, accuracy against tuned CC at two candidate-set rates; a bar joins the two rates of a dataset.

$p_{\text{Holm}} = 0.016$ for both), was at least as accurate as the strongest PLL baseline on 10/10 and 9/10 datasets, and stayed within 0.8 and 1.9 accuracy points of the oracle [C8]. The advantage holds where weak labels are hardest: at $q = 0.6$ In-Context Gibbs is +0.205 accuracy ahead of the naive TFM, and the four datasets with at least eight classes show no collapse. Removing datasets flagged for group structure or drift changes no conclusion (Appendix D).

6.3 Where the gain comes from: sampling the labels jointly

With the same TFM and the same warm start, In-Context Gibbs had lower log loss than hard-EM, which assigns each latent label its most likely value, in 78.9% of 38 cells in Study 2 and 86.7% of 128 cells across all six weak-label settings of Study 1; the posterior probability that it is better was 1.00 in both, and the advantage grew with the weakness of the labels (Spearman $\rho = 0.66$; Appendix F) [C7]. The gain over the naive TFM follows Proposition 2: large for candidate sets and bag counts, absent where the naive labels already carry all the information (Appendix I) [C12].

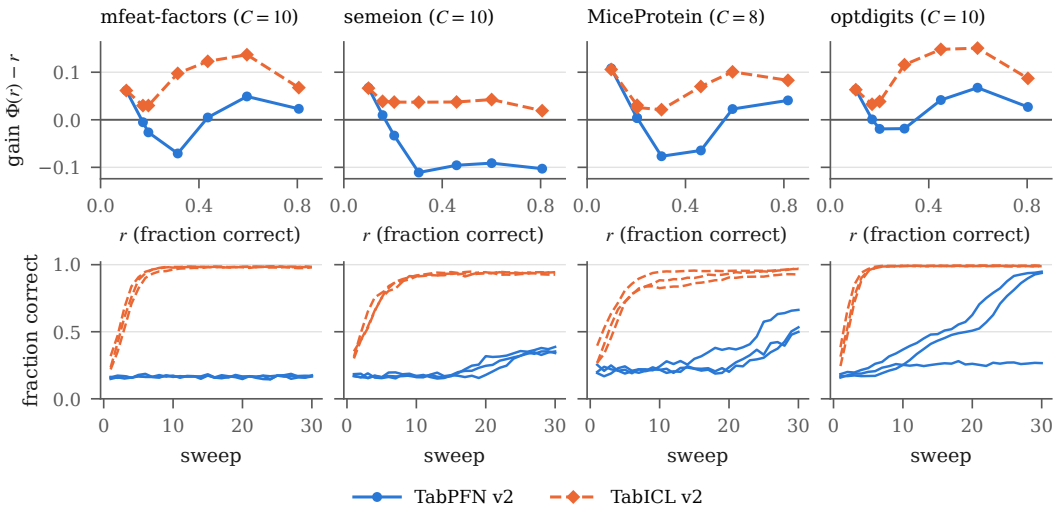


Figure 2: PLL at $q = 0.6$ on the four Study 1 datasets with at least eight classes. Top: gain $\Phi(r) - r$ of one Gibbs sweep started from an assignment with a fraction r of correct labels (mean over 3 seeds). Bottom: fraction of correct training labels along the chains of the main runs, one line per seed (diagnostic only; the sampler never sees the true labels).

6.4 What the gain depends on: the TFM's response to noisy context

Study 1, with TabPFN v2, isolated the condition (Table 2, left). In-Context Gibbs again improved substantially on the naive TFM, by +0.148 AUC on LLP and +0.099 and +0.180 accuracy on PLL, and led the primary comparators by +0.024 AUC (8 of 10 datasets) and +0.022 accuracy at $q = 0.3$, short of the pre-registered criteria. At $q = 0.6$ it was ahead on ten datasets by +0.033 on average (7 of 10), while on the four with at least eight classes the chains stalled near an uninformative labelling (Figure 2, bottom), which turned the average into a deficit of -0.132.

The stalling is a property of the TFM, not of the sampler. Figure 2 (top) measures what one sweep does to an assignment in which a fraction r of the labels is correct. With TabPFN v2 the gain $\Phi(r) - r$ turns negative for intermediate r on all four datasets: noisy context labels are reinforced, and a chain that starts near chance is pulled back to it. With TabICL v2 the gain is positive everywhere, and its chains label 93–99% of the training points correctly within 5–15 sweeps. In the registered TFM ablation on the same datasets, the sampler with TabICL v2 did not stall (lowest test accuracy on the four datasets 0.94) and passed every pre-registered test (Appendix C). Because that TFM was selected after seeing these results, Study 2 tested it prospectively, with the outcome above [C13]. The response map is cheap and needs only labelled development data, so it could be used to screen a TFM before use. [TODO: link to the collapse-threshold analysis (G9_PLAN D2), if a proposition is ready by 20/12]

6.5 Real weak labels

On three real partial-label datasets (BirdSong, Lost, MSRCv2; 10 random 90/10 splits; TabICL v2, registered in Study 1 for datasets with more than ten classes), In-Context Gibbs had the highest mean accuracy on all three and the best mean rank among 10 methods (2.15; Table 15) [C9]. It led the best baseline by +0.028 on BirdSong and +0.045 on MSRCv2; on Lost (+0.002) the top three were within 0.003.

6.6 Exactness, mixing and cost

The exact sampler does not predict better. The MH-corrected block sampler, exact when the TFM is (Proposition 3), did not predict systematically better than the default at the same 30 sweeps (Table 12): with TabPFN v2, +0.010 AUC on LLP and +0.006 accuracy on PLL; with TabICL v2, +0.033 AUC on LLP (with worse log loss), and -0.009, -0.021 and -0.016 accuracy on PLL at $q = 0.3$, 0.6 and on datasets with $\geq$8 classes. It accepted as few as 0.12 of proposals and was a median 7 times slower. The exact transductive predictor (Theorem 1(c)) was likewise no better (-0.013 AUC on LLP), and K, the number of sweeps or the initialisation changed results by at most 0.015 (Appendix F).

Mixing. Two chains per cell disagree on a noticeable fraction of individual labels (median share with $\widehat{R} > 1.1$ [30] between 6% and 31%), yet their test metrics differ by a median of at most 0.041 on the three smallest datasets of each group (Table 12). Each seed of Study 2 runs its own chain, so the confirmatory tests include this Monte Carlo variation.

Cost. No weight is trained. On one T4 GPU a run took a median of 41 s (maximum 5.3 min) with TabICL v2 for up to 7,000 training points, using at most 3.7 GiB of GPU memory [C11].

7 Limitations

Dependence on the TFM. The method inherits its TFM's response to noisy context labels: with TabPFN v2 it stalled under heavy ambiguity with many classes. The two TFMs were compared on the same data only in Study 1 and in the response-map analysis, so the cross-study contrast is not a controlled comparison of TFMs. **Theory versus the default sampler.** Our guarantees cover the single-site, swap and MH-corrected samplers; the uncorrected block samplers used by default have none that is non-vacuous. The exact sampler did not predict better overall, but it mixes more slowly, so the bias of the approximation has not been isolated on real data. Tables have at most 10,000 rows, and contexts above 2,000 points are subsampled (14 of 28 datasets in Study 2), an approximation Theorem 2 covers only through $\bar{\varepsilon}$, not isolated. **Mixing.** Thirty sweeps give stable predictions, not draws from a converged posterior over individual labels. **Scope.** The claims cover LLP with bags of 8 and PLL. As Proposition 2 predicts, the sampler did not improve on the naive TFM for PU with labels selected completely at random, class-dependent noise or semi-supervised learning, and it was not significantly better than the primary comparator for bags of 16 or 32 (Study 1). On PLL the gain over tuned CC is about one accuracy point, close to the oracle ceiling.

8 Conclusion

Under exchangeability, a frozen tabular foundation model's predictive is the full conditional of a Gibbs sampler over unknown labels. In-Context Gibbs turns this into one training-free procedure for candidate sets and label proportions, with exact variants for an exact TFM and stability under an approximate one, and in a pre-registered study on new datasets it outperformed the strongest specialised learners for both. Because it inherits its TFM's response to noisy context labels, progress in TFMs carries over directly. Open directions are a theory of the stalling threshold, guarantees for block samplers, and weak labels whose observation process depends on the features.

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A Proofs

Assumptions. (A0) $W \sim p(W \mid X, y)$, independent of (x_*, y_*) given (X, y); nuisance parameters of the observation model, if any, are a priori independent of (ϕ, X, y). (A1) $q(\cdot \mid x, D) = p(\cdot \mid x, D)$ for every context the algorithm queries. (A2) $p(k \mid x, D) > 0$ for all k and every context of positive probability. (A3) ℓ is known and $\sum_y p(y \mid X)\ell(y) > 0$. For instance-level observations $\ell(y) = \prod_i \ell_i(y_i)$; the set of free points is $F = \{i : |\{k : \ell_i(k) > 0\}| \geq 2\}$.

A.1 Lemma 1

Proof of Lemma 1. Given ϕ the pairs (x_j, y_j) are i.i.d., hence exchangeable, so the joint law of $(D_{-i}, (x_i, y_i))$ equals that of $(D_{n-1}, (x_*, y_*))$, a context of $n-1$ points and a query. The information (X, y_{-i}) coincides with (x_i, D_{-i}). Therefore

$$p(y_i \mid y_{-i}, X) = \frac{p((x_i, y_i), D_{-i})}{\sum_k p((x_i, k), D_{-i})} = p(y_i \mid x_i, D_{-i}). \qquad \square$$

The PPD conditions on the query features x_* as well, which may carry information about ϕ; this is the quantity the PFN training loss targets, since x_* is drawn from the prior together with the context [22].

Corollary 1 (Positivity)**.** *Under (A2),* $p(y \mid X) > 0$ *for every* $y \in [C]^n$.

Proof. Some y^0 has $p(y^0 \mid X) > 0$. Change one coordinate at a time, $y^0 \to y^1 \to \cdots \to y^m = y$, step k changing point s_k. If $p(y^{k-1} \mid X) > 0$, then $p(y^{k-1}_{-s_k} \mid X) > 0$, the context $D^{k-1}_{-s_k}$ has positive probability, and by Lemma 1 and (A2) $p(y^k \mid X) = p(y^{k-1}_{-s_k} \mid X)\, p(y^k_{s_k} \mid x_{s_k}, D^{k-1}_{-s_k}) > 0$. Induct on k. $\square$

A.2 Theorem 1

Proof of Theorem 1(a). The update of point i is $P_i(y, y') = \pi_i(y'_i \mid y_{-i}) \mathbb{1}[y'_{-i} = y_{-i}]$ with $\pi_i(k \mid y_{-i}) \propto q(k \mid x_i, D_{-i})\ell_i(k)$. By Lemma 1 and (A1), $\pi_i(k \mid y_{-i}) \propto p(k \mid y_{-i}, X)\ell_i(k) \propto \pi(k, y_{-i})$, the full conditional of π; hence $\pi(y)P_i(y, y') = \pi(y_{-i})\pi_i(y_i \mid y_{-i})\pi_i(y'_i \mid y_{-i})$ is symmetric in (y, y'). By Corollary 1, $\pi(y) > 0$ iff $\ell(y) > 0$, a product set; any two of its points are joined by changing one coordinate at a time inside the set, each step with probability at least $\pi_i(\cdot \mid \cdot)/|F| > 0$, so the random-scan kernel $P = |F|^{-1} \sum_{i \in F} P_i$ is irreducible, and aperiodic since $P(y, y) > 0$. A finite, irreducible, aperiodic chain converges geometrically. The implementation sweeps the free points in a uniformly random order, $P_{\mathrm{sw}} = \frac{1}{|F|!} \sum_\sigma P_{\sigma(1)} \cdots P_{\sigma(|F|)}$; each factor preserves π, and

P_{sw} is reversible because the adjoint of a product is the product in reverse order and the set of permutations is closed under reversal. □

Proof of Theorem 1(b). Choose a bag uniformly; if it is not pure, choose uniformly an ordered pair (i, j) in it with $y_i \neq y_j$ and propose y' by swapping their labels. Brook's lemma through the intermediate state (y'_i, y_{-i}) gives

$$\frac{p(y' \mid X)}{p(y \mid X)} = \frac{p(y'_i \mid y_{-i}, X)}{p(y_i \mid y_{-i}, X)} \cdot \frac{p(y'_j \mid y'_i, y_{-ij}, X)}{p(y_j \mid y'_i, y_{-ij}, X)},$$

and Lemma 1 with (A1) turns the right-hand side into the ratio R_{ij} of four TFM predictives that the algorithm computes with two forward passes. A swap preserves the counts, so $\ell(y') = \ell(y)$ and $R_{ij} = \pi(y')/\pi(y)$, independently of the order of (i, j). The number of differently labelled pairs depends only on the counts, so the proposal is symmetric and the chain is Metropolis with stationary distribution π. Two assignments of a bag with the same counts are joined by a sequence of swaps preserving the counts, each with positive probability by Corollary 1; bags are independent, so the product chain is irreducible (the lazy version is also aperiodic). □

Proof of Theorem 1(c). (Inductive.) The ergodic theorem for a finite irreducible chain applied to the bounded function $y \mapsto p(y_* \mid x_*, (X, y))$, with (A1), gives almost sure convergence of (3) to $\bar{p}$. By the law of total probability and (A0), $p(y_* \mid x_*, X, W) = \sum_y p(y \mid x_*, X, W)\, p(y_* \mid x_*, (X, y))$; the difference of the expectations of one $[0, 1]$-valued function under two measures is at most their TV distance. If the prior is separable, $\phi = (\phi_x, \phi_{y|x})$ independent a priori with $p(x \mid \phi) = p(x \mid \phi_x)$ and $p(y \mid x, \phi) = p(y \mid x, \phi_{y|x})$, then $p(y \mid x_*, X, W) \propto \ell(y) \int p(y \mid X, \phi_{y|x}) p(\phi_{y|x}) d\phi_{y|x}$ times a factor that does not depend on y, so $p(y \mid x_*, X, W) = p(y \mid X, W)$ and the bound is zero. (Transductive.) Adding x_* as a free point with $\ell_* \equiv 1$ gives a chain for $p(y, y_* \mid X, x_*, W)$ that is exact by part (a) (for LLP, the swap chain combined with single-site updates of x_*, each preserving the target). By Lemma 1, the Rao–Blackwell term is $q(y_* \mid x_*, (X, y^{(s)}))$, whose average converges to $p(y_* \mid x_*, X, W)$. □

Proof of Proposition 1. By the tower property, $\mathbb{E}[S(f, y_*)] = \mathbb{E}\big[\mathbb{E}[S(f, y_*) \mid X, W, x_*]\big]$. For each value of (X, W, x_*), f is a fixed distribution and $y_* \sim p(\cdot \mid x_*, X, W)$; strict propriety gives the unique minimiser $f = p(\cdot \mid x_*, X, W)$. Its risk is finite (at most $\log C$ for the log score). □

A.3 Theorem 2

Proof of Theorem 2. Existence and uniqueness. The state space is finite, so $\tilde{P}$ has a stationary distribution. With $q > 0$ every single-site update and every MH acceptance probability is positive, so $\tilde{P}$ has the transition graph of P and is irreducible; $\tilde{\pi}$ is unique. Compatibility of the conditionals of q is never used. *Perturbation identity.* Let $\Delta = \tilde{\pi} - \pi$ and $\nu = \tilde{\pi}(\tilde{P} - P)$, so $\nu(\Omega) = 0$. From $\tilde{\pi} = \tilde{\pi}\tilde{P}$ and $\pi = \pi P$, $\Delta = \nu + \Delta P$, and iterating, $\Delta = \sum_{t<T} \nu P^t + \Delta P^T$. *Contraction.* For a signed measure μ with $\mu(\Omega) = 0$, $\|\mu K\|_1 \leq \delta(K)\|\mu\|_1$. If $\tau(P) < \infty$ then $\delta(P^T) \to 0$, the remainder vanishes, and $\|\Delta\|_1 \leq \|\nu\|_1 \tau(P)$; if $\tau(P) = \infty$ there is nothing to prove (no aperiodicity is needed). Finally $\|\nu\|_1 \leq \sum_y \tilde{\pi}(y) \|\tilde{P}(y, \cdot) - P(y, \cdot)\|_1 = 2\bar{\varepsilon}$. *Single-site error.* $\tilde{P}(y, \cdot) - P(y, \cdot)$ averages the differences of the single-site updates, and each is the TV distance between $q\ell_i$ and $p\ell_i$ after normalisation. With $Z_p = \sum_k p(k)\ell_i(k)$, $2\,\mathrm{TV}(q_\ell, p_\ell) \leq \big(\sum_k \ell_i(k)|q(k) - p(k)| + |Z_q - Z_p|\big)/Z_p \leq 4(\max_k \ell_i(k)/Z_p)\,\mathrm{TV}(q, p)$, so the error is at most $2\kappa\varepsilon_{\mathrm{TV}}$ with $\kappa = \sup \max_k \ell_i(k)/Z_p$. *Prediction.* $|\mathbb{E}_{\tilde{\pi}} q - \mathbb{E}_\pi p| \leq |\mathbb{E}_{\tilde{\pi}}(q - p)| + |\mathbb{E}_{\tilde{\pi}} p - \mathbb{E}_\pi p| \leq \varepsilon'_{\mathrm{TV}} + \mathrm{TV}(\tilde{\pi}, \pi)$. □

Relation to classical bounds. With $E = \tilde{P} - P$ and the fundamental matrix Z, $\tilde{\pi} - \pi = \nu Z$ [21, 25], which gives $\mathrm{TV}(\tilde{\pi}, \pi) \leq \tau_1(Z)\bar{\varepsilon}$ with $\tau_1(Z) \leq \tau(P)$ [7, 26]; (4) is the weaker, more interpretable form. It is attained by a two-state chain. For a single-site sweep the per-sweep error is at most $|F|$ times the per-update error; the MH swap kernel has error at most $4\varepsilon_\infty$, with $\varepsilon_\infty = \sup|\log q - \log p|$.

A.4 Block updates

Proposition 3 (Blocks)**.** *(a) A block update whose proposal is the product of block-wise TFM predictives (for LLP, conditioned on bag counts), accepted with the MH ratio computed by Brook's lemma along the changed labels, leaves π invariant when $q = p$. (b) Without the MH step, Theorem 2 applies with an error $\bar{\varepsilon}_B$ that is bounded only under an extra stability assumption on the PPD, and the bound grows with n for fixed K.*

Proof of Proposition 3. (a) The proposal Q_B depends only on y_{-B}, which the step does not change, so the update is MH-within-Gibbs with an independent proposal. Brook's lemma along the changed labels with Lemma 1 gives the exact ratio $p(y' \mid X)/p(y \mid X)$, multiplied by the ratio of ℓ (one for LLP). The proposal is positive on the support of the block and acceptance is positive, so the chain is irreducible. (b) Let P_{bG} be exact block Gibbs. Under the stability assumption that adding one labelled point or one unlabelled feature to a context changes the PPD by at most β in TV, the chain rule and a hybrid argument give $\mathrm{TV}\big(\prod_{i\in B\cap F} p(\cdot \mid x_i, D_{-B}),\, p(y_{B\cap F} \mid y_{-(B\cap F)}, X)\big) \le m_f(m-1)\beta$ for a block of m points of which m_f are free, and after reweighting by ℓ, $\bar{\varepsilon}_B \le 2\kappa_B\, m_f(m-1)\beta$; Theorem 2 with $P = P_{\mathrm{bG}}$ concludes. □

Why this is vacuous at our sizes. With $\beta = O(1/n)$ and fixed K ($m = n/K$), $m_f(m-1)\beta = O(n/K^2)$ grows with n. For bag-blocks, κ_B is the inverse probability that all bags of a group match their counts under the product proposal: for $n \approx 2{,}000$ and $K = 10$, $\log_{10}\kappa_B \approx 15.8$ for bags of 8. On an exactly solvable LLP problem (3 bags of 4), the uncorrected bag-block chain has $\mathrm{TV}(\tilde{\pi}, \pi)$ between 0 and about 0.6 depending on the prior and grouping, so the approximation is real at small n; at our sizes we measure its effect on predictions instead (Section 6.6).

A.5 When the naive TFM suffices

Complement to Proposition 2(a). For binary labels with class prior ϖ, for any score s, $\mathrm{AUC}(\eta) - \mathrm{AUC}(s) \le \mathbb{E}|s - \tilde{\eta}|/(\lambda\varpi(1-\varpi))$: the AUC regret of a learner fitted to naive labels is its own estimation error, amplified by $1/\lambda$.

Proof of Proposition 2 and of the complement. (a) With $P(\tilde{w} = k \mid x, y) = M_{y,k}$ independent of x and $M_{y,k} = b_k$ for $y \neq k$, $\tilde{\eta}_k = M_{k,k}\eta_k + b_k(1-\eta_k) = b_k + \lambda_k\eta_k$, $\lambda_k = M_{k,k} - b_k$. For PLL with each wrong class entering the candidate set independently with probability q and the naive label chosen uniformly from the set, $M_{y,y} = a = \mathbb{E}[1/(1+N)] = \frac{1-(1-q)^C}{Cq}$ with $N \sim \mathrm{Bin}(C-1, q)$ and $b = (1-a)/(C-1)$, so $\lambda = (Ca-1)/(C-1) > 0$ for $q < 1$. For LLP with random bags of size k, $P(\tilde{w}_i = 1 \mid x_i, y_i) = (y_i + (k-1)\varpi)/k$, so $\lambda = 1/k$. A strictly increasing affine map preserves ranks, hence one-vs-rest AUC. For the binary bound, write $\varpi(1-\varpi)\mathrm{AUC}(s) = \mathbb{E}[\eta(X)(1-\eta(X'))h_s(X,X')]$ for independent copies, with $h_s = \mathbb{1}[s(X) > s(X')] + \frac{1}{2}\mathbb{1}[s(X) = s(X')]$; symmetrising, $\varpi(1-\varpi)[\mathrm{AUC}(\eta) - \mathrm{AUC}(s)] = \frac{1}{2}\mathbb{E}\big[|\Delta|(\mathbb{1}[s \text{ reverses the pair}] + \frac{1}{2}\mathbb{1}[s \text{ ties}, \Delta \neq 0])\big]$ with $\Delta = \eta(X) - \eta(X')$. With $t = (s-b)/\lambda$, which has the ranks of s, a reversed or tied pair has $|\Delta| \le |\eta(X) - t(X)| + |\eta(X') - t(X')|$, so the right-hand side is at most $\mathbb{E}|s - \tilde{\eta}|/\lambda$. For $\lambda = 1$ and scores that are class-probability estimates, this is Corollary 12 of Agarwal [1]. (b) $p_\theta(\tilde{w}) = \sum_O p_\theta(O)R(\tilde{w} \mid O)$ gives $\partial_\theta \log p_\theta(\tilde{w}) = \mathbb{E}_\theta[\partial_\theta \log p_\theta(O) \mid \tilde{w}]$, and Jensen's inequality for conditional expectations gives $I_{\tilde{w}}(\theta) \le I_O(\theta)$, with equality iff the score of O is $\sigma(\tilde{w})$-measurable. □

Information lost by the naive labels (logistic model $\eta = \sigma(\theta x)$, $\theta = 2$, $x \sim \mathcal{N}(0, I)$; per point). LLP: the Godambe information of the pointwise (composite) likelihood on naive labels, relative to the information of the bag counts, is 0.134, 0.060, 0.033, 0.017, 0.009 for bags of 2, 4, 8, 16, 32. PLL: the naive labels retain 0.24–0.26 of the information in the candidate sets at $q = 0.3$ and 0.06–0.10 at $q = 0.6$ ($C \in \{3, 4, 7\}$). For PU-SCAR and label noise the ratio is 1.

B Pre-registration and timeline

Both studies were run under a written protocol with hypotheses, comparators, tests, datasets and decision rules fixed before any evaluation data were used; every later change was logged as a dated deviation. In the anonymised repository each freeze is a commit, and every result file records the hashes of the code that produced it.

Table 3: Timeline of the two studies (2026, UTC). "Exploratory" marks analyses that were not part of the registered tests.

Date	Event
26 Oct	Study 1 protocol frozen (hypotheses H1–H5, datasets, baselines, tuning spaces, primary-comparator rule, tests, ablations, sensitivity analysis). Later changes were logged as dated deviations: all but three before the evaluation runs; the three made during the analysis (dropping an external LLP benchmark whose data were unavailable, fixing the reading of an ambiguous rule to its weakest form, labelling the TFM ablation exploratory) are reported.
28 Oct–1 Nov	Study 1 runs on the evaluation datasets (TabPFN v2; registered TabICL v2 ablation; baselines).
30 Oct	Study 1 analysis: H2 and H4 not met, H3 and H5 met; TFM ablation passes all tests (exploratory). Decision to run a confirmatory replication changing only the TFM.
1 Nov	Study 2 dataset census and selection from metadata only; flags for the sensitivity analysis set by hand.
1 Nov 15:19	**Study 2 frozen** (commit), before any Study 2 table was downloaded.
1–2 Nov	Data download and post-check rule applied (two datasets fail $n \geq 500$ after de-duplication; first reserve added, subsampled to 10,000 rows). Exploratory response-map runs on Study 1 datasets only.
2–3 Nov	Study 2 runs; confirmatory analysis with the pre-written script. One post-hoc fix of a floating-point tie in the analysis (a tie counted as a loss); no conclusion changed. Failed cells of one baseline rerun once.
3 Nov	Exploratory runs on Study 1 datasets: $\widehat{R}$ and MH-corrected blocks with TabICL v2.

C Study 1 with TabICL v2 (registered TFM ablation)

Table 4 applies the H2 and H4 tests of Study 1 to the registered TFM ablation, in which every TFM-based method uses TabICL v2. With clean labels the two TFMs are about equally accurate (oracle within 0.005); the difference appears only with weak labels. Because TabICL v2 was promoted after these results were seen, they are exploratory; Study 2 is the confirmatory test.

Table 4: Study 1 datasets, TabICL v2 for all TFM-based methods (exploratory). Same tests and comparators as Table 2.

Setting	Comparator	Δ [95% CI]	W/T/L	p_{Holm}	mean: In-Context Gibbs (TabICL / TabPFN)	/ oracle (TabICL)	
LLP, bag 8	DLLP (linear)	+0.058 [+0.029, +0.093]	10/0/0	0.004	0.889	0.855	0.928
PLL, $q = 0.3$	CC, tuned	+0.029 [+0.013, +0.051]	12/0/2	0.004	0.854	0.847	0.865
PLL, $q = 0.6$	CC, tuned	+0.038 [+0.013, +0.071]	12/0/2	0.009	0.831	0.661	0.865

D Study 2: sensitivity analysis, all baselines and per-dataset results

Sensitivity. Five datasets were flagged before any data were downloaded (repeated measurements of the same units, or temporal drift): SpeedDating and eye_movements (binary); JapaneseVowels, cjs and gas-drift (multiclass). Without them, every conclusion is unchanged (Table 5). With seven multiclass datasets, Holm-adjusted Wilcoxon tests cannot reach 0.05 unless every comparison is won; the registered rule for this case checks that the mean difference keeps its sign and that In-Context Gibbs still wins on at least 60% of the remaining datasets, which holds at both levels.

Table 5: Study 2 without the flagged datasets.

Setting	Comparator	datasets	Δ (all datasets)	Δ (flagged removed)	W/T/L	p_{Holm}	Δ vs naive
LLP, bag 8	DLLP (linear)	16	+0.052	+0.051	15/0/1	$<$0.001	+0.124
PLL, $q = 0.3$	CC, tuned	7	+0.012	+0.016	6/1/0	0.062	+0.023
PLL, $q = 0.6$	CC, tuned	7	+0.012	+0.013	5/1/1	0.062	+0.168

All baselines. Against each baseline separately (Holm over all baseline–level pairs of a setting), In-Context Gibbs is significantly better than all ten LLP baselines; on PLL every mean difference is positive and 7 of 16 pairs are significant after this stricter correction (Table 6). One baseline (default PRODEN) produced non-finite probabilities on 5 of 6 gas-drift cells, also on a rerun; its means use the remaining cells.

Table 6: Study 2, In-Context Gibbs against every baseline. Bold: significant in favour of In-Context Gibbs.

Setting	Baseline	datasets	Δ	W/T/L	p_{Holm}
LLP, bag 8 (AUC)	DLLP (MLP)	18	+0.057	17/0/1	$<$**0.001**
	DLLP (MLP), tuned	18	+0.041	16/0/2	$<$**0.001**
	DLLP (linear)	18	+0.052	16/0/2	$<$**0.001**
	DLLP (linear), tuned	18	+0.046	16/0/2	$<$**0.001**
	LLP-VAT	18	+0.062	16/0/2	$<$**0.001**
	LLP-VAT, tuned	18	+0.042	16/0/2	$<$**0.001**
	EasyLLP	18	+0.116	18/0/0	$<$**0.001**
	EasyLLP, tuned	18	+0.069	17/0/1	$<$**0.001**
	GLWS	18	+0.056	17/1/0	$<$**0.001**
	GLWS, tuned	18	+0.051	18/0/0	$<$**0.001**
PLL, $q = 0.3$ (acc.)	PRODEN	9	+0.048	9/0/0	**0.043**
	PRODEN, tuned	10	+0.027	8/1/1	0.070
	GLWS/ILL	10	+0.048	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.014	10/0/0	**0.031**
	RC	10	+0.034	9/0/1	**0.043**
	RC, tuned	10	+0.013	8/1/1	0.070
	CC	10	+0.035	10/0/0	**0.031**
	CC, tuned	10	+0.012	8/2/0	0.070
PLL, $q = 0.6$ (acc.)	PRODEN	10	+0.173	10/0/0	**0.031**
	PRODEN, tuned	10	+0.031	8/0/2	0.070
	GLWS/ILL	10	+0.128	10/0/0	**0.031**
	GLWS/ILL, tuned	10	+0.015	8/0/2	0.070
	RC	10	+0.048	8/0/2	0.070
	RC, tuned	10	+0.012	8/1/1	0.070
	CC	10	+0.047	8/0/2	0.070
	CC, tuned	10	+0.012	8/1/1	0.070

Table 7: Study 2, LLP with bags of 8: AUC per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

Dataset	Gibbs	hard-EM	naive	DLLP-lin	DLLP tuned	LLP-VAT tuned	oracle
ada	**0.898**	0.895	0.742	0.888	0.880	0.883	0.912
Bank_Customer_Churn	0.849	**0.855**	0.785	0.750	0.837	0.834	0.877
compas-two-years	**0.683**	0.671	0.539	0.677	0.650	0.662	0.692
eye_movements	0.564	**0.589**	0.538	0.569	0.549	0.548	0.701
Fitness_Club	**0.831**	0.784	0.609	0.743	0.719	0.753	0.831
hazelnut-contaminant	**0.993**	0.989	0.853	0.941	0.953	0.951	0.995
heloc	**0.797**	0.788	0.759	0.789	0.788	0.787	0.813
Is-this-a-good-customer	0.633	0.634	0.576	0.610	**0.656**	0.631	0.737
jasmine	**0.870**	0.841	0.780	0.779	0.807	0.792	0.902
Marketing_Campaign	**0.878**	0.870	0.667	0.609	0.630	0.635	0.915
mushroom	**1.000**	**1.000**	0.721	1.000	0.996	1.000	1.000
NATICUSdroid	0.983	**0.985**	0.911	0.973	0.976	0.977	0.988
philippine	0.774	**0.781**	0.711	0.681	0.699	0.700	0.934
profb	0.580	0.495	0.507	0.578	0.569	**0.587**	0.675
scene	0.971	**0.983**	0.823	0.973	0.977	0.973	0.991
socmob	**0.980**	0.979	0.794	0.956	0.950	0.947	0.987
SpeedDating	**0.842**	0.752	0.768	0.708	0.783	0.745	0.871
sylvine	0.993	**0.996**	0.955	0.960	0.965	0.958	0.997
mean	0.840	0.827	0.724	0.788	0.799	0.798	0.879

Table 8: Study 2, PLL: accuracy per dataset (mean over 3 seeds). Bold: best non-oracle method among those shown.

	$q = 0.3$						$q = 0.6$					
Dataset	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle	Gibbs	hard-EM	naive	CC tuned	RC tuned	oracle
anneal	**0.991**	0.989	0.955	0.976	0.972	0.996	**0.977**	0.945	0.801	0.957	0.962	0.996
cardiotocography	**1.000**	**1.000**	0.994	**1.000**	**1.000**	1.000	**1.000**	0.886	0.476	**1.000**	**1.000**	1.000
cjs	0.997	0.997	0.962	0.997	**0.998**	1.000	**0.993**	0.830	0.391	0.991	0.989	1.000
gas-drift	0.996	**0.997**	0.993	0.991	0.992	0.998	**0.995**	0.994	0.930	0.988	0.986	0.998
JapaneseVowels	0.991	**0.995**	0.968	0.984	0.980	0.998	**0.989**	0.983	0.785	0.970	0.973	0.998
MIC	**0.887**	0.886	0.883	0.873	0.872	0.891	**0.886**	0.877	0.848	0.870	0.871	0.891
students-dropout	0.777	**0.782**	0.775	0.754	0.754	0.786	**0.770**	0.764	0.744	0.734	0.733	0.786
website_phishing	**0.823**	0.820	0.792	0.787	0.787	0.844	0.786	**0.787**	0.735	0.778	0.778	0.844
wine-quality-white	0.563	**0.571**	0.538	0.547	0.546	0.585	**0.556**	0.510	0.481	0.539	0.537	0.585
yeast	**0.597**	0.578	0.543	0.588	0.591	0.606	0.561	0.456	0.275	**0.563**	0.563	0.606
mean	0.862	0.862	0.840	0.850	0.849	0.870	0.851	0.803	0.647	0.839	0.839	0.870

Table 9: Study 2: mean score of every method (datasets averaged over 3 seeds). PRODEN (default) failed on gas-drift in 5 of 6 cells (non-finite probabilities, reproduced on a rerun); its $q = 0.3$ mean is over 9 datasets and its $q = 0.6$ gas-drift entry uses one seed.

LLP, bag 8 (AUC, 18 datasets)			PLL (accuracy, 10 datasets)				
Method	default	tuned	Method	$q = 0.3$		$q = 0.6$	
				default	tuned	default	tuned
In-Context Gibbs (ours)	0.840		**In-Context Gibbs (ours)**	0.862		0.851	
Hard-EM (same TFM)	0.827		Hard-EM (same TFM)	0.862		0.803	
Stochastic EM (same TFM)	0.835		Stochastic EM (same TFM)	0.858		0.843	
Naive TFM	0.724		Naive TFM	0.840		0.647	
DLLP (MLP)	0.783	0.799	PRODEN	0.800	0.836	0.678	0.820
DLLP (linear)	0.788	0.794	GLWS	0.814	0.848	0.724	0.837
LLP-VAT	0.778	0.798	RC	0.828	0.849	0.803	0.839
EasyLLP	0.724	0.771	CC	0.827	0.850	0.804	0.839
GLWS	0.784	0.789	Oracle (true labels)	0.870		0.870	
Oracle (true labels)	0.879						

E Correctness experiments

We enumerate every labelling of small problems ($n \leq 14$), build the exact transition matrix of each kernel with an analytic PPD as q, and compare its stationary distribution with π. Two priors are used: one in which features carry no information about the label mechanism, and a three-component mixture in which they do. Across all 14 configurations the TV distance is at most 3×10^{-14} (Table 10), and running the actual sampler for 10^5 sweeps reaches the Monte Carlo floor of i.i.d. sampling. For Theorem 2 we use conditionals with deliberate, incompatible perturbations (30 configurations) and small PFNs trained on the same prior; the bound held in 42 of 42 configurations and was 1.9 to 15 times loose (median 3.9; Table 11); a fully trained PFN gives TV 0.012 (single-site) and 0.007 (swap). On an exactly solvable pooled problem with $n = 1{,}000$ and $K = 10$, blocking over all points changes predictions by 1.7×10^{-4}, whereas blocking only the free points biases them by 0.031, which is why blocks include fixed points.

Table 10: Exactness (Theorem 1, Proposition 3(a)): TV between the stationary distribution of the exact kernel and π.

Prior	Weak labels	Kernel	n	states	$\mathrm{TV}(\text{stationary}, \pi)$
cell	PU-SCAR (c=0.5)	single	10	128	3.3e-15
cell	SSL	single	10	128	4.0e-15
cell	PLL	single	10	128	4.0e-15
cell	Noisy (T known)	single	8	256	1.9e-14
cell	LLP (3 bags of 4)	swap	12	96	3.1e-15
cell	SSL, block-MH (m=3)	block-mh	10	128	2.8e-14
mixture	PU-SCAR (c=0.5)	single	10	128	3.3e-15
mixture	SSL	single	10	128	3.0e-15
mixture	PLL	single	10	128	3.0e-15
mixture	Noisy (T known)	single	8	256	2.3e-14
mixture	LLP (3 bags of 4)	swap	12	96	4.8e-15
mixture	SSL, block-MH (m=3)	block-mh	10	128	4.0e-15
cell	PLL (C=3, |S_i|=2)	single	7	128	1.2e-14
mixture	PLL (C=3, |S_i|=2)	single	7	128	3.8e-15

Table 11: Stability (Theorem 2): actual TV between the stationary distributions of the approximate and exact chains, the bound $\tau(P)\bar{\varepsilon}$, and their ratio (looseness).

Conditionals	Kernel	configs	median $\mathrm{TV}(\tilde{\pi}, \pi)$	median bound	bound held	median looseness
perturbed PPD (incompatible)	single	15	0.0181	0.246	15/15	9.8
perturbed PPD (incompatible)	swap	15	0.0254	0.066	15/15	2.2
small PFN (trained)	single	6	0.1419	0.628	6/6	4.9
small PFN (trained)	swap	6	0.0480	0.107	6/6	3.3

F Ablations

Table 12: Mixing and the exact sampler, by TFM. TabPFN: Study 1 (registered ablations); TabICL: exploratory lot on the Study 1 datasets. "3 small": the three smallest datasets of each group, as registered for $\widehat{R}$. Columns 2–5: medians over cells of the fraction of training labels with $\widehat{R} > 1.1$ (two chains) and of the absolute difference between the test metrics of the two chains. Columns 6–7: mean difference in AUC (LLP) or accuracy (PLL) between the MH-corrected and the default sampler at 30 sweeps, with the mean MH acceptance rate. In the last row, $\widehat{R}$ uses all four datasets with $\geq$8 classes, and the MH comparison only two of them (mfeat-factors, optdigits; 5 of 12 cells), because the corrected sampler ran out of the 12-hour compute session on optdigits.

	labels with $\widehat{R} > 1.1$		test-metric gap, 2 chains		MH-corrected − default	
Setting (datasets)	TabPFN	TabICL	TabPFN	TabICL	TabPFN	TabICL [acc. rate]
LLP, bag 8 (3 small)	11%	28%	0.041	0.018	+0.010	+0.033 [0.68]
PLL, $q = 0.3$ (3 small)	6%	6%	0.006	0.018	+0.006	−0.009 [0.78]
PLL, $q = 0.6$ (3 small)	21%	31%	0.016	0.021	–	−0.021 [0.72]
PLL, $q = 0.6$, $\geq$8 classes (4; MH: 2)	–	22%	–	0.002	–	−0.016 [0.47]

Table 13: In-Context Gibbs versus hard-EM with the same TFM and warm start, log loss. A cell is a (dataset, weak-label level) pair; Bayesian signed-rank test with ROPE ± 0.005; Spearman correlation between weakness level and the gap. †Computed on PLL cells (LLP has a single level), as stated before the analysis was run.

Study (TFM; settings)	cells	Gibbs better	P(Gibbs better)	P(ROPE)	Spearman ρ
Study 1 (TabPFN v2; all six settings)	128	86.7%	1.00	0.00	0.36
Study 2 (TabICL v2; LLP bag 8, PLL)	38	78.9%	1.00	0.00	0.66†

Table 14: Study 1 ablations (TabPFN v2): each variant against the default on the same cells. Blocks, sweeps and MH: the three smallest datasets per group; transductive and cold start: all datasets.

Variant (vs. default)	Setting	cells	Δ AUC/acc.	W/T/L	Δ log loss (+ = better)
$K = 20$	LLP, bag 8	9	+0.007	5/0/4	+0.024
$K = 20$	PLL, 0.3	9	+0.006	5/2/2	−0.000
$K = 5$ (default 10)	LLP, bag 8	9	−0.005	6/0/3	+0.006
$K = 5$ (default 10)	PLL, 0.3	9	+0.004	5/1/3	+0.009
15 sweeps (default 30)	LLP, bag 8	9	+0.005	4/0/5	+0.001
15 sweeps (default 30)	PLL, 0.3	9	+0.004	4/2/3	−0.001
60 sweeps	LLP, bag 8	9	+0.014	4/0/5	−0.012
60 sweeps	PLL, 0.3	9	+0.005	4/5/0	+0.009
MH-corrected blocks (exact)	LLP, bag 8	9	+0.010	4/0/5	−0.019
MH-corrected blocks (exact)	PLL, 0.3	9	+0.006	4/3/2	+0.009
transductive prediction	LLP, bag 8	30	−0.013	7/0/23	−0.042
cold start	LLP, bag 8	30	−0.005	10/0/20	−0.002
cold start	PLL, 0.3	42	+0.000	11/23/8	+0.000
cold start	PLL, 0.6	42	+0.012	14/9/19	+0.023
PU-SAR prior $N(0, 1)$	PU-SAR, 0.1	30	−0.002	15/3/12	−0.015
PU-SAR prior $N(0, 1)$	PU-SAR, 0.3	30	+0.001	10/1/19	+0.002
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.1	30	−0.004	7/9/14	+0.000
PU-SAR prior $N(0, 4^2)$	PU-SAR, 0.3	30	−0.001	10/12/8	−0.016

G Real partial labels

Table 15: Real partial labels (Study 1, TabICL v2 for all TFM-based methods): accuracy averaged over 10 random 90/10 splits, and mean rank over all splits of the three datasets among the ten ranked methods (In-Context Gibbs, naive TFM and eight baselines). Bold: best non-oracle accuracy.

Method	BirdSong	Lost	MSRCv2	mean rank
In-Context Gibbs (ours)	**0.759**	**0.750**	**0.628**	2.15
Hard-EM (same TFM)	0.730	0.665	0.627	–
Stochastic EM (same TFM)	0.744	0.716	0.622	–
Naive TFM	0.689	0.481	0.491	7.27
PRODEN (default)	0.667	0.373	0.344	9.88
PRODEN (tuned)	0.707	0.748	0.465	5.12
GLWS (default)	0.722	0.712	0.582	3.98
GLWS (tuned)	0.731	0.747	0.490	4.03
RC (default)	0.719	0.720	0.424	6.00
RC (tuned)	0.702	0.721	0.478	5.23
CC (default)	0.719	0.717	0.425	6.02
CC (tuned)	0.700	0.723	0.479	5.32
Oracle (true labels)	0.835	0.889	0.824	–

H Datasets, baselines and tuning

Datasets. Study 1 uses every OpenML-CC18 dataset with real (non-synthetic) data, 500 to 10,000 rows, at most 500 features and, for binary tasks, a minority class of at least 10% (multiclass: 3–10 classes), one per source; Study 2 applies the same criteria to five curated suites after excluding OpenML-CC18 and Study 1 (Table 17). Duplicate rows are removed before splitting. TFMs receive ordinally encoded features; MLP baselines receive median-imputed, standardised features with one-hot encoded categories, fitted on the training split.

Baselines and tuning. Each baseline runs in the configuration of its original code or paper and tuned by 20 random configurations (the default among them) from the spaces below, scored by the method's own loss on a 20% validation split with weak labels only (split by bag for LLP); the best configuration is refit on the full training split. MLP-based learners (DLLP, LLP-VAT, EasyLLP, GLWS, PRODEN) search learning rate $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$, weight decay $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$, hidden layers in $\{(), (64), (128, 128)\}$ and epochs in $\{100, 200, 400\}$, plus bags per batch for LLP, VAT parameters for LLP-VAT, the optimiser for PRODEN and the entropy term for GLWS/ILL; RC and CC use a linear model with the PRODEN space (Table 16).

Table 16: Search spaces (20 random configurations, the default included). LogU: log-uniform.

Baseline	Default	Search space
DLLP (MLP, linear), EasyLLP	lr 10^{-3}, wd 0, 200 epochs	lr $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$; wd $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$; hidden $\in \{(), (64), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; bags/batch $\in \{8, 32\}$
LLP-VAT	as DLLP + VAT defaults of the official code	as DLLP; $\xi \in \{10^{-6}, 10^{-3}\}$; $\epsilon \in \{0.1, 0.5, 1, 2\}$; consistency weight $\sim \mathrm{LogU}(10^{-2}, 1)$
GLWS / ILL	official code, MLP backbone	lr, wd as DLLP; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$; entropy term on/off (PLL)
PRODEN	lr 10^{-3}, wd 10^{-5}, hidden (128, 128), 200 epochs, SGD	lr, wd, hidden, epochs as DLLP; optimiser $\in$ {SGD (momentum 0.9), Adam}
RC, CC	linear model as in the original paper	as PRODEN
SAR-EM (PU-SAR)	$C_f = C_e = 1$	$C_f, C_e \sim \mathrm{LogU}(10^{-3}, 10^{3})$
nnPU, uPU, forward correction	lr 10^{-3}, wd 0, hidden (100), 200 epochs	lr $\sim \mathrm{LogU}(10^{-4}, 10^{-2})$; wd $\sim \mathrm{LogU}(10^{-6}, 10^{-2})$; hidden $\in \{(64), (128), (128, 128)\}$; epochs $\in \{100, 200, 400\}$
VPU	official code	lr $\sim \mathrm{LogU}(10^{-5}, 10^{-3})$; mixup $\lambda \in \{0.01, 0.03, 0.1, 0.3\}$; $\alpha \in \{0.1, 0.3, 1\}$
Self-training	logistic regression, $C = 1$, threshold 0.75	$C \sim \mathrm{LogU}(10^{-3}, 10^{3})$; threshold $\in \{0.6, 0.75, 0.9, 0.95\}$
Label spreading	kNN, $k = 7$, $\alpha = 0.2$	kernel $\in$ {kNN, RBF}; $k \in \{3, 5, 7, 11, 15, 21\}$; $\gamma \sim \mathrm{LogU}(10^{-3}, 10)$; $\alpha \in \{0.1, 0.2, 0.5, 0.8\}$
VIME	official code	$p_m \in \{0.1, 0.3, 0.5\}$; $\alpha \in \{0.5, 1, 2\}$; $\beta \in \{0.1, 1, 10\}$; hidden $\in \{(64), (128)\}$
Confident learning + TFM	default (not tuned)	–

Table 17: Datasets (n after removing duplicates; *subsampled to 10,000 rows by the registered rule; S1: flagged for the sensitivity analysis).

Study 1 (OpenML-CC18)						Study 2 (new curated suites)					
ID	name	n	d	C	S1	ID	name	n	d	C	S1
3	kr-vs-kp	3,196	36	2		24	mushroom	8,124	22	2	
29	credit-approval	690	15	2		312	scene	2,400	299	2	
44	spambase	4,210	57	2		470	profb	672	9	2	
50	tic-tac-toe	958	9	2		934	socmob	1,156	5	2	
1049	pc4	1,344	37	2		40536	SpeedDating	8,378	120	2	✓
1489	phoneme	5,395	5	2		41143	jasmine	2,984	144	2	
1494	qsar-biodeg	1,052	41	2		41145	philippine	5,832	308	2	
6332	cylinder-bands	540	37	2		41146	sylvine	5,124	20	2	
23381	dresses-sales	500	12	2		41156	ada	4,146	48	2	
38	sick	3,711	29	2		44130	eye_movements	7,608	20	2	✓
11	balance-scale	625	4	3	✓	45026	heloc	9,444	22	2	
12	mfeat-factors	1,994	216	10		45039	compas-two-years	2,712	11	2	
28	optdigits	5,620	64	10		46911	Bank_Customer_Churn	10,000	10	2	
46	splice	3,006	60	3		46927	Fitness_Club	1,499	6	2	
182	satimage	6,430	36	6		46930	hazelnut-spread-contamin	2,400	30	2	
188	eucalyptus	736	19	5		46938	Is-this-a-good-customer	1,723	13	2	
458	analcatdata_authorship	841	70	4		46940	Marketing_Campaign	2,058	25	2	
469	analcatdata_dmft	556	4	6		46969	NATICUSdroid	7,491	86	2	
1475	first-order-theorem-prov	5,636	51	6		181	yeast	1,453	8	10	
1497	wall-robot-navigation	5,456	24	4	✓	375	JapaneseVowels	9,961	14	9	✓
1501	semeion	1,593	256	10		1466	cardiotocography	2,115	35	10	
40966	MiceProtein	1,080	77	8	✓	23380	cjs	1,811	33	6	✓
40975	car	1,728	6	4	✓	40498	wine-quality-white	3,961	11	7	
40982	steel-plates-fault	1,941	27	7		46906	anneal	886	38	5	
						46960	students_dropout_and_	4,424	36	3	
						46963	website_phishing	724	9	3	
						46980	MIC	1,699	111	8	
						1476	gas-drift	10,000*	128	6	✓

I Secondary settings of Study 1

Study 1 also registered secondary settings (Table 18). In-Context Gibbs improved strongly on the naive TFM for larger bags, as for bags of 8, but not for PU with labels selected completely at random, class-dependent noise or SSL, where its difference to the naive TFM was at most -0.013 AUC. This is the pattern Proposition 2 predicts, and it was registered in advance as exploratory analyses D1–D3: the advantage over the naive TFM grows from $q = 0.3$ to $q = 0.6$ (D1, one-sided $p = 0.039$) and from bags of 8 to 16 (D2, $p = 0.010$), and stays below $+0.02$ AUC in the settings where the naive labels lose no information (D3, all six levels). Against the primary comparators of these settings no difference was significant except class-dependent noise at $r = 0.4$, where confident learning combined with the TFM was better.

Table 18: Study 1 secondary settings (TabPFN v2). Δ positive favours In-Context Gibbs. †Log loss after matching the known class prior (sign flipped so that positive is better).

Setting	Metric	Primary comparator	In-Context Gibbs − comparator			In-Context Gibbs − naive TFM	
			Δ	W/T/L	p_{Holm}	Δ	W/T/L
LLP, bag 16	AUC	DLLP (linear)	+0.040	7/0/3	0.232	+0.206	9/0/1
LLP, bag 32	AUC	DLLP (linear)	+0.048	6/0/4	0.375	+0.223	10/0/0
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.1$	AUC	nnPU	+0.034	6/0/4	0.967	+0.007	6/0/4
PU-SAR, $\mathbb{E}[e \mid y = 1] = 0.3$	AUC	nnPU	+0.032	8/0/2	0.109	+0.017	6/0/4
PU-SCAR, $c = 0.1$	log loss†	VPU, tuned	+0.118	8/0/2	0.111	−0.033	5/0/5
PU-SCAR, $c = 0.3$	log loss†	VPU, tuned	+0.179	8/0/2	0.111	−0.028	2/0/8
Class-dependent noise, $r = 0.2$	AUC	CL + TFM	−0.006	3/0/7	0.105	−0.013	1/0/9
Class-dependent noise, $r = 0.4$	AUC	CL + TFM	−0.048	2/0/8	0.020	−0.054	1/0/9
SSL, 5% labelled	AUC	Self-training, tuned	−0.002	5/0/5	1.000	−0.041	0/0/10
SSL, 20% labelled	AUC	Self-training, tuned	−0.009	8/0/2	0.750	−0.016	0/0/10

J Statements

Data and code availability. All datasets are public (OpenML; the PALM collection for real partial labels). Code, frozen protocols, raw per-cell results with code and data hashes, and the scripts that generate every number, table and figure of this paper are available in an anonymised repository and will be released.

Compute. All runs used free Kaggle sessions with two NVIDIA T4 GPUs (TFM-based methods) or CPUs (baselines). Study 1 used about 42 hours of two-GPU sessions and 26 hours of four-process CPU sessions; Study 2 about 4 GPU-hours and 80 CPU process-hours (two sessions of about ten hours with four processes each); the exploratory lots about 13 hours of two-GPU sessions. Development and pilot runs before each freeze are not included.

Broader impacts. Learning from weak labels lowers the cost of annotation and makes coarse or ambiguous labels usable. Learning from label proportions also recovers individual-level predictions from data released only in aggregate; when aggregation was chosen to protect individuals, such models can weaken that protection, and releases of aggregated data should be assessed with this in mind. We have not measured how much individual-level information the sampler recovers from bag counts. [TODO: quantify, or keep this sentence (G9/REVIEW_RESPONSE_PLAN R8)]

Use of large language models. An AI assistant was used for writing code, drafting proofs that the authors checked, running analyses through the pre-written scripts, and drafting text. The authors designed the studies, approved every pre-registration and decision, ran all experiments and are responsible for all content. [TODO: align with the venue's LLM policy; docs/AI_USAGE.md]

NeurIPS Paper Checklist

1. **Claims.** Do the main claims in the abstract and introduction accurately reflect the paper's contributions and scope? **[Yes]** Every claim is traced to a theorem or a pre-registered test (Sections 5 and 6); the scope (LLP with bags of 8 and PLL; dependence on the TFM) is stated in the abstract, Section 6.4 and Section 7.

2. **Limitations. [Yes]** Section 7, including the unmet hypotheses of Study 1, the lack of a non-vacuous guarantee for the default sampler, incomplete label-level mixing and the size of the PLL gains.

3. **Theory assumptions and proofs. [Yes]** Assumptions (A0)–(A3) in Section 3; complete proofs in Appendix A, numerically checked in Appendix E.

4. **Experimental result reproducibility. [Yes]** Protocols, splits, weak-label generators, tuning spaces and all settings are specified (Section 6.1, Appendix H); every number is generated by scripts from raw per-cell results.

5. **Open access to data and code. [Yes]** Public datasets; anonymised code and raw results (Appendix J).

6. **Experimental setting/details. [Yes]** Sections 4 and 6.1, Appendix H.

7. **Experiment statistical significance. [Yes]** Wilcoxon signed-rank tests with Holm correction, bootstrap confidence intervals, win/tie/loss counts, Bayesian signed-rank tests, and a registered sensitivity analysis.

8. **Experiments compute resources. [Yes]** Section 6.6 (cost per run) and Appendix J.

9. **Code of ethics. [Yes]** The research uses public data and conforms to the NeurIPS Code of Ethics.

10. **Broader impacts. [Yes]** Learning from weak labels lowers the cost of annotation. Learning from label proportions also recovers individual-level predictions from data released only in aggregate, which can weaken the privacy that aggregation was meant to provide (Appendix J).

11. **Safeguards. [NA]** No model or data with high risk of misuse is released.

12. **Licenses for existing assets. [Pending: licences of the TabICL v2 checkpoint, each OpenML dataset and the PALM partial-label data not yet confirmed]** Draft answer, to be confirmed asset by asset: TabPFN v2 weights: Prior Labs License (Apache 2.0 with an attribution requirement); TabICL code: BSD-3-Clause; datasets: licences as listed on OpenML; baseline code without a licence is loaded at run time and not redistributed.

13. **New assets. [Yes]** Code and raw results, documented in the repository.

14. **Crowdsourcing and research with human subjects. [NA]**

15. **IRB approvals. [NA]**

16. **Declaration of LLM usage. [Yes]** Appendix J.